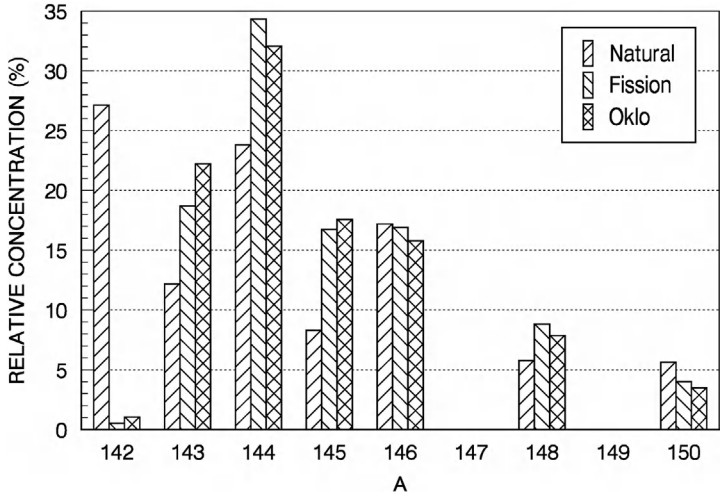


Figure 19.11 Comparison of the isotopic composition of neodymium from normal Nd ore, from ^{235}U fission in a BWR and from Oklo.

periodically as the heat from fission boiled away the water so that the chain reaction ceased after a while. When water returned (as the temperature decreased) the chain reaction would resume. The neutron flux probably never exceeded 10^{13} n m^{-2} s^{-1}.[5] The fluence is estimated to exceed 1.5×10^{25} n m^{-2}. This would have consumed about 6 t ^{235}U, releasing a total energy of 2−3 GWy, at a power level probably $\leq$10 kW. About 1.0−1.5 t of ^{239}Pu was formed by neutron capture in ^{238}U, but the relatively short (on a geological scale) half-life of ^{239}Pu allowed it to decay into ^{235}U. Since a few samples enriched in ^{235}U also have been found, it is believed that in some places in Oklo breeding conditions may temporarily have existed, cf. §22.12.

It has been calculated that many uranium rich ore deposits $2-3 \times 10^9$ y ago must have been supercritical in the presence of moderating water. Therefore natural nuclear chain reactions may have had an important local influence on the early environment of earth.

19.11. REACTOR CONCEPTS

Nuclear reactors are designed for production of heat, mechanical and electric power, radioactive nuclides, weapons material, research in nuclear physics and chemistry, etc. The design depends on the purposes, e.g. in the case of electric power production the design is chosen to provide the cheapest electricity taking long term reliability in consideration. This may be modified by the availability and economy of national resources such as raw material, manpower and skill, safety reasons, etc. Also the risk for proliferation of reactor

[5] n-fluxes and fluencies are given per m^2 in the SI system; however, it is still common to use per cm^2.

materials for weapons use may influence the choice of reactor type. Many dozens of varying reactor concepts have been formulated, so we must limit the discussion in this chapter to a summary of the main variables, and the most common research and power reactors. Fast reactors and some other designs are discussed in Chapter 20.

We have already mentioned three basic principles for reactor design: the neutron energy (thermal or fast reactors), the core configuration (homogeneous or heterogeneous aggregation of fuel and coolant), and the fuel utilization (burner, converter or breeder). In the homogeneous reactor the core can be molten metal, molten salt, an aqueous or an organic solution. In heterogeneous reactors the fuel is mostly rods filled with metal oxide. The fuel material can be almost any combination of fissile and fertile atoms in a mixture or separated as in the core (fissile) and blanket (fertile) concept (Ch. 20). The choice of moderator is great: H_2O, D_2O, Be, graphite, or organic liquid. There is even more choice in the coolant which can be molten metal, molten salt, liquid H_2O, D_2O, or organic liquid, as well as gaseous CO_2, helium or steam.

Some of the more important combinations are summarized in Tables 19.1 and 20.1.

19.12. RESEARCH AND TEST REACTORS

According to the World Nuclear Industry Handbook 2012, 264 research and test reactors were in operation throughout the world at the end of 2011, but some have probably been shut down since then. While the smaller research reactors may have an operating power $\leq 1\ kW_{th}$, and do not need forced cooling, the larger test reactors operate at $\leq 50\ MW_{th}$. Many of these reactors have facilities for commercial radionuclide production.

Research reactors are used in nuclear physics, in nuclear, analytical and structural chemistry, in radiobiology, in medicine, etc. They are usually easy to operate, inherently safe, and of moderate cost. Many of them are of the pool type; the reactor core is located in the center of a stainless steel clad concrete vessel, 6−8 m deep and 2−5 m wide, containing purified water. The water provides the main radiation shielding, moderation, and cooling. The concrete walls are about 2 m thick when the reactor is located above ground; otherwise much thinner walls can be used. The main radiation protection demand comes from the reaction

$$^{16}O(n,p)^{16}N \xrightarrow[7.13\ s]{\beta\gamma} {}^{16}O \tag{19.39}$$

since the γ's emitted are very energetic (6.1 MeV). At power levels $>100\ kW_{th}$ forced cooling may be required. The fuel contains usually highly enriched uranium (20 to 90% ^{235}U) as rods or plates; the amount of ^{235}U is ≤ 3 kg.

Some reactors are designed to produce very high neutron fluxes, either for testing materials (especially for fast reactors) or for isotope production. The high flux reactor at Grenoble, France, has fuel plates of highly enriched (93%) UAl_3 canned in aluminum.